Lecture 16

Saturday, April 11, 2026 9:21 AM

Today: $\underbrace{\text{Logistic Regression}}_{\text{Classification}}$

Given a data point → output label

Examples:

- Spam filter: Is the email spam $(y=1)$ or not $(y=0)$
- Medical diagnosis: Does the patient have this condition or not
- Fraud detection: Is this transaction fraudulant or not.

Training data

$(x^{(i)}, y^{(i)}) \quad 1 \leq i \leq N$

$y^{(i)} \in \{0, 1\}$

$x^{(i)} \in \mathbb{R}^d$ (d features)

Examples:

① $x^{(i)}$: systolic blood pressure reading
$y^{(i)}$: chronic high blood pressure present?

② "buy" "click" "free" "unsubscribe" $d = 10000$
frequency of the i^{th} word in the email

$y^{(i)} = \begin{cases} 1 & \text{spam} \\ 0 & \text{not spam} \end{cases}$

Logistic regression:

Solves this problem using a hypothesis function.

$h_\theta(x)$ $\quad x \in \mathbb{R}^d, \quad \theta \in \mathbb{R}^d$

θ → parameter

x → feature vector for a datapoint

$h_\theta(x)$: measures $\mathbb{P}(y=1 \mid x; \theta)$

x → data point; θ → parameter

$h_\theta(x) : \mathbb{R}^d \to [0,1]$

1. $0 \leq h_\theta(x) \leq 1$ $\quad (h_\theta(x) = p(y=1 \mid x; \theta))$
2. x's score : $\theta^T x$ $\quad (\in \mathbb{R})$
   - a larger value of $\theta^T x \Rightarrow h_\theta(x) \to 1$
   - a smaller value of $\theta^T x \Rightarrow h_\theta(x) \to 0$
3. $h_\theta(x) = 1/2$ decision threshold
   - $h_\theta(x) < 1/2 \quad \hat{y} = 0$
   - $h_\theta(x) > 1/2 \quad \hat{y} = 1$

What is form of $h_\theta(x)$?

Candidate 1 : $h_\theta(x) = \theta^T x$ ← does not satisfy property 1

$0 \leq h_\theta(x) \leq 1$

Instead of probability, look at odd

The odds of this stock going up are 3-to-1

⇔ " probability " " " " " is 75% or 0.75

$$\text{odds} = \frac{h_\theta(x)}{1 - h_\theta(x)} \in (0, \infty)$$

Candidate 2 : $\frac{h_\theta(x)}{1 - h_\theta(x)} = \theta^T x$ ← does not satisfy property 1

Take logs

$$\log \text{odds} = \log \left(\frac{h_\theta(x)}{1 - h_\theta(x)} \right) \in (-\infty, \infty)$$

Logistic regression assumes:

$$\boxed{\log\left(\frac{h_\theta(x)}{1-h_\theta(x)}\right) = \theta^T x} \quad \text{①}$$

θ → parameters

x → data

Event E

$\mathbb{P}(E)$: probability of event E

odd (E) : $\dfrac{\mathbb{P}(E)}{1-\mathbb{P}(E)}$

From ① :

$$\frac{h_\theta(x)}{1-h_\theta(x)} = \exp(\theta^T x)$$

$$\Rightarrow \quad h_\theta(x)\left[1+\exp(\theta^T x)\right] = \exp(\theta^T x)$$

$$\Rightarrow \quad h_\theta(x) = \frac{1}{1+\exp(-\theta^T x)} \quad \text{②}$$

$$\mathbb{P}(y=1 \mid x;\theta) = h_\theta(x) = \frac{1}{1+\exp(-\theta^T x)} = g(\theta^T x)$$

$$\left.g(z) = \frac{1}{1+e^{-z}}\right\} \text{ sigmoid function}$$

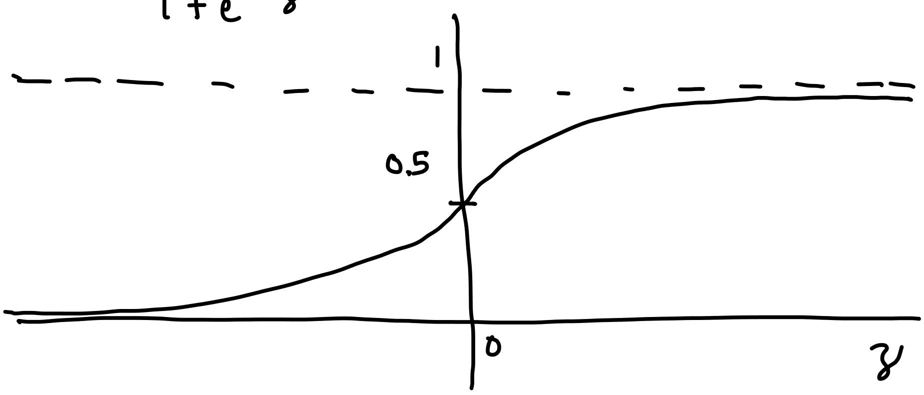
tl;dr

$$\mathbb{P}(y=1 \mid x;\theta) = h_\theta(x) = \frac{1}{1+\exp(-\theta^T x)}$$

Modeling assumption of logistic regression

Properties of the sigmoid function

$$g(z) = \frac{1}{1+e^{-z}}$$



① $g'(z) = g(z)(1-g(z))$
② $1-g(z) = g(-z)$
$\}$ will be important later

Summary:

Logistic regression assumes

$$h_\theta(x) = \underbrace{g}_{\text{Sigmoid function}}(\theta^T x) = \frac{1}{1+e^{-\theta^T x}} = \mathbb{P}(y=1 \mid x;\theta)$$

Q1: How to decide the value of θ?
Q2: How to compute $\underset{\in\{0,1\}}{\hat{y}}$ from $h_\theta(x)$?

Ans 2: θ is known
a new sample x comes, $\hat{y} = ?$

$$h_\theta(x) = \begin{cases} > 1/2 & \hat{y} = 1 \\ < 1/2 & \hat{y} = 0 \\ = 1/2 & \text{toss a fair coin} \end{cases} \quad \Bigg\} \text{Rule 1}$$

$$h_\theta(x) = \frac{1}{1+\exp(-\theta^T x)} = \mathbb{P}(y=1 \mid x;\theta)$$

$$h_\theta(x) > \frac{1}{2} \Rightarrow \theta^T x > 0$$

$$\left(\frac{1}{2} < \frac{1}{1+\exp(-\theta^T x)} \right.$$
$$\Rightarrow 1+\exp(-\theta^T x) < 2$$
$$\Rightarrow \exp(-\theta^T x) < 1$$
$$\Rightarrow -\theta^T x < 0$$
$$\left. \Rightarrow \theta^T x > 0 \right)$$

$$h_\theta(x) < \frac{1}{2} \Rightarrow \theta^T x < 0$$

Rule 1 $\equiv$ Rule 2 : Predict $\hat{y}=1$ if $\theta^T x > 0$
Predict $\hat{y}=0$ if $\theta^T x < 0$
Toss a coin if $\theta^T x = 0$

Back to Q1 : How to compute θ ?

Assumptions :

1. Training data samples are i.i.d.
2. $\mathbb{P}(y=1 \mid x;\theta) = h_\theta(x) = g(\theta^T x) = \dfrac{1}{1+\exp(-\theta^T x)}$
3. Classes are Bernoulli random variables

$$y \mid x;\theta \sim \text{Ber}(h_\theta(x))$$

$$\Rightarrow \quad y \mid x;\theta = \begin{cases} 1 & \text{w.p. } h_\theta(x) \\ 0 & \text{w.p. } 1-h_\theta(x) \end{cases}$$

Likelihood function

$L(\theta)$: measure of what is the probability of seeing our training data given θ.

$$\rightarrow L(\theta) = \prod_{i=1}^{N} \mathbb{P}(y^{(i)} \mid x^{(i)};\theta) \quad \text{— (3)}$$

$$\rightarrow \theta^* = \underset{\theta}{\text{argmax}} \; L(\theta) \longleftarrow$$ a principled way to obtained θ^*

$$\left.\begin{aligned} \mathbb{P}(y^{(i)}=1 \mid x^{(i)};\theta) &= h_\theta(x^{(i)}) \quad \text{— (4)} \\ \mathbb{P}(y^{(i)}=0 \mid x^{(i)};\theta) &= 1-h_\theta(x^{(i)}) \quad \text{— (5)} \end{aligned}\right\} \text{modelling assumption}$$

$$\Rightarrow \mathbb{P}(y^{(i)} \mid x^{(i)};\theta) = h_\theta(x^{(i)})^{y^{(i)}} \left(1-h_\theta(x^{(i)})\right)^{(1-y^{(i)})} \quad \text{— (6)}$$

(4) & (5) $\Longleftrightarrow$ (6) . Using (3) & (6)

$$L(\theta) = \prod_{i=1} h_\theta(x^{(i)})^{y^{(i)}} \left(1-h_\theta(x^{(i)})\right)^{(1-y^{(i)})} \quad (7)$$

$$\theta^* = \arg\max_\theta L(\theta)$$

Some manipulations to make our life simple, log of (7)

$$\log L(\theta) = \sum_{i=1}^{N} y^{(i)} \log h_\theta(x^{(i)}) + (1-y^{(i)}) \log(1-h_\theta(x^{(i)}))$$

maximize $\log L(\theta)$

$\Rightarrow$ minimizing $-\log L(\theta) = J(\theta)$

$$J(\theta) = -\sum_{i=1}^{N} y^{(i)} \log h_\theta(x^{(i)}) + (1-y^{(i)}) \log(1-h_\theta(x^{(i)}))$$

1. $J(\theta)$ is convex
2. $J(\theta)$ is called the cross-entropy function

$$\theta^* = \arg\min_\theta J(\theta)$$

Gradient descent or any of its variants can be used.

$$\theta \leftarrow \theta - \frac{\alpha}{N} X^T (g(X^T\theta) - y)$$